

MAINS MASTER PROGRAM (MMP) 2024

ENVIRONMENT-6

FINAL UPDATES

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2. CLIMATE CHANGE ISSUES

1) METHANE

- **Practice Questions:**
 - » Discuss the sources, implications, and potential mitigation strategies of methane emissions in the context of global warming and climate change. How can international cooperation play a significant role in addressing this significant environmental concern? [15 marks, 250 words]
- **Introduction:**
 - » As per UNEP, Methane is a GHG which is responsible for 30% of the warming since pre-industrial times. It's contribution is 2nd only to carbondioxide.
- **Why special focus on methane is needed in our fight against climate change?**
 - » Methane has much higher global warming potential than CO₂.
 - » IPCC had said that the methane mitigation has the greatest potential to slow warming over the next 20 years.
- **Methane Emission: Sources:**
 - » **Natural Sources:** Wetlands, termites etc.
 - **Wetlands** are the largest source of methane.
 - » **Agriculture** - Rice cultivation, animal husbandry etc. generate substantial amount of methane.
 - » **Energy Production** (fossil fuel) - Among anthropogenic factors, after Agriculture, it is this sector which contributes to the highest methane production. It is released during the extraction, processing, and transport of fossil fuels, including coal, oil, and natural gas.
 - » **Leakage:** For e.g. the ruptures in the underwater Nord stream in Sep 2022 caused the single largest such release of the greenhouse gas.
 - » **Landfills** in recent times are also becoming a big source of methane emissions.
 - » **Thawing of permafrost** in polar region is also releasing methane. In future, it may become a big source of methane emissions.
- **Steps being taken:**
 - » **International Steps:**
 - **Improving Detection:** UNEP has launched International Methane Emissions observatory - the Methane Alert and Response System (MARS) at COP27. It is focused on scaling up global efforts to detect and act on major emissions sources in a transparent manner and accelerate implementation of the global methane pledge.
 - **Methane Reduction: Global Methane Pledge** announced at COP26
 - By COP27, 150 countries have joined the initiative lead by USA and EU. They have promised to cut their methane emission by at least 30% from 2020 levels by 2030.
 - **Why has India not joined the pledge?**
 - India's methane emissions are 'survival emissions' and not 'luxury emissions'.
 - The two prominent sources of methane in India are enteric fermentation and 'paddy cultivation' and any restriction on them would harm small and marginal farmers.
 - Other than harming farmers, it may also reduce agri production. Currently, India is one of the largest producers and exporters of rice.

- India also argues that 6th IPCC report has highlighted that CO₂ is the major global warming gas and this pledge is shifting focus to methane which has a lifetime of only 12 years, whereas CO₂ can survive for more than 100 years.
- **India has not joined the global methane pledge**, but it doesn't mean the India is not worried about methane emissions. There are several fronts on which India is working.
 - National Innovation in Climate Resilient Agriculture (NICRA) project of ICAR has developed several technologies with the potential to mitigate methane emissions.
 - For instance, the 'System of Rice Intensification' has the potential to enhance rice yield from 36-49% with 22-35% less water than conventional transplanted rice. It also uses less seed, fertilizers, and pesticides.
 - Another technology, 'Direct Seeded Rice' reduces methane emissions as it does not involve raising nurseries, puddling, and transplanting.
 - **Harit Dhara**: It is an anti-methanogenic feed supplement developed by ICAR. It can cut down cattle methane emissions by 17-20% and can also result in higher milk production.
 - Under Crop Diversification Program, methane emission is being avoided due to diversion of paddy to alternate crops like pulses, oilseeds, maize, cotton, and agro-forestry.
- **Way Forward:**
 - » **Renewable Energy Transition:** In long run it will reduce dependency on fossil fuels which will reduce emissions of both CO₂ and methane.
 - » **Alternate Agricultural practices:**
 - Improving the effectiveness and yield of rice cultivation methods like System of Rice Intensification and Direct Seeded Rice and encouraging more farmers to adopt these practices.
 - Crop diversification to reduce dependency on rice.
 - » **Focus on Burp Control:**
 - Promote anti-methanogenic feed supplement like **Harit Dhara**.
 - More R&D on alternatives. For e.g. in 2021 EU approved a food supplement, Bovaer, saying that it can consistently reduce methane emissions from dairy cows by 30-80%.
 - » **Scientific Waste Management:** Reduce the waste disposal on landfills; ensure installation of landfill gas capture systems etc.; converting organic waste into biogas which can be used for energy etc.
 - » **Leak Detection and Repair:** Regular monitoring and maintenance of oil and gas infrastructure can minimize methane leaks.
 - » **Improved International Cooperation:** Global targets; data sharing, finance mobilization; technology transfers; Improved R&D are some of the methods by which international cooperation can contribute in fighting the challenge of methane.
- **Conclusion:** Addressing methane emissions is critical for mitigating global warming and its associated impacts. A comprehensive approach dealing with fossil fuel sector, agriculture sector and international cooperation will be needed for a more resilient and climate-resilient future.

2) IMPACT OF CLIMATE CHANGE

GWG emissions is breaching all the records: As per the AR6, **Emissions of Carbon dioxide, methane and nitrous oxide breached records in 2020**. CO2 Concentration in the atmosphere - at around 419 parts per million - are the highest they have been in 2 million years.

- **Three factors** make carbon budgeting complex:
 1. **The pollutants** - primarily GHGs like CO2 and methane - have an extraordinary long life. Thus historic emissions continue to warm up the planet just like current emissions.
 2. GHG emissions are linked to economic growth.
 3. **Sharing of burden** becomes difficult as the emissions are associated with economic growth.
- 1. **GWG emissions** is breaching all the records: As per the AR6, **Emissions of Carbon dioxide, methane and nitrous oxide breached records in 2020**. CO2 Concentration in the atmosphere - at around 419 parts per million - are the highest they have been in 2 million years.
- 2. **Rising Temperatures**
 - As per the AR6 of IPCC, the global temperature has already risen by 1.1 degree C since preindustrial 19th century. This could increase upto 1.5 degree Celsius in less than 20 years (before 2040).
- 3. **Melting of Glaciers and Sea Level Rise -> Submergence of coastal region**
 - AR6: Sea level rise has tripled compared with 1901-1971. The Antarctic Sea ice is the lowest in last 1,00 years.
 - The temperature of Antarctica rose above 20 degree Celsius for the first time on record.
- 4. **Heating up of Oceans -> marine heat waves, intense cyclones etc.**
 - Ocean temperatures are better indicators of global warming as 93% of the excess solar energy trapped by GHG accumulate in the world's oceans (AR5 report in 2014).
- 5. **Increasing variability in weather patterns**
 - **Heat waves and floods** which used to be once-in-a century event are becoming more regular occurrence.
 - **Weather Disasters** have displaced millions of people this year and **affected rainfall patterns** from India to northern Russia and the Central United states.
- 6. **Compounding extremes** (several climate change drivers operating together) are maximizing disaster in India and elsewhere.
 - E.g. heavy rainfall, landslides, snow avalanches, and flooding occurring together is an example of compounding event.
- 7. **Thawing of Permafrost and Arctic Lakes Bubbling Methane**
 - **Warmer temperature increases the thawing of permafrost and release methane to the atmosphere**
 - **Presently Arctic is a net carbon sink**
 - **But**, soon arctic could become a carbon source, if the earth continues to warm, and a lot of permafrost thaws out.

8. **Sea Water** is **26% more acidic** than at the start of the industrial era. This is leading to degradation of marine ecosystem.
9. **Biodiversity Loss**
- At least **1 million species were at risk** because of the rising CO₂ concentration in the atmosphere and global warming.
 - For instance, a recent study shows that seal pups (IUCN: LC) are finding it tough to survive in the Baltics in the absence of ice. 100s of grey seal pups are dying on the shores of the Baltic Sea in Estonia and Latvia as the Nordic coastline faced winter without ice in decades.
10. **Negative Impact on Food Security, Agriculture and Economy**
- Variability in rainfall
 - Increased temperature and evaporation of water sources
 - Increased chances of Locust attacks
 - Extreme weather events triggered by climate change costs India \$87 billion annually : State of Climate in Asia, 2020 (report by WMO)
11. Climate change has adversely affected **both physical and mental health of people**.
- Extreme heat events -> Mortality and morbidity
 - Climate related food borne and water borne diseases has increased. The incidence of vector borne diseases have also increased due to range expansion and/or increased reproduction of disease vectors.
 - Some mental health challenges are associated with increasing temperatures, trauma from weather and climate extreme events, and loss of livelihood and culture. Exposure to wildfire smoke, atmospheric dust, and aeroallergens have been associated with climate sensitive cardiovascular and respiratory distress.
12. **Achievements of SDG targets** have been negatively hindered.
13. **Impact on Cities, Settlements and Infrastructure**
- Hot extremes including heatwaves have intensified in cities. This has also aggravated air pollution events and limited functioning of key infrastructure. Infrastructure, including transportation, water, sanitation and energy systems have been compromised by extreme and slow onset of events.
14. **Economic Impacts**
- Damages due to variability in weather pattern (e.g. slow onset of Monsoons) and extreme weather events.
 - Climate exposed sectors like agriculture, forestry, fishery, energy and tourism have been adversely affected. Outdoor labour productivity have gone down
 - Extreme events like cyclones hinder economic growth in short run too.
15. **Shrinking of Stratosphere:**
- According to a study published by NASA, the earth's energy imbalance doubled over the 14 year period between 2005 - 2019, doubling the pace at which the Earth retains heat from 2005. As a result of this **we are already on the brink of losing stratosphere**.

16. **Climate Migration**

- **Introduction**

- As sea level rise, temperature becomes unbearable, and disasters more severe, tens of millions of people may not be able to stay where they are and will have to migrate. This is climate migration (or Climate Change induced migration). This has already started and is putting people at risk.

- **Impact:**

- **Displacement** (especially of the most vulnerable groups)
 - **IPCC's 6th AR** says that through displacement and involuntary migration from extreme weather and climate events, climate change has generated and perpetuated vulnerability.
 - Since 2008, an average of more than 20 million people per year have been displaced by extreme weather events, many of which were exacerbated by climate change.
- **Human toll**
- **Disruption in Economic and Political Stability -> Conflicts**

3) PARIS AGREEMENT AT UNFCCC COP21

- **The Paris Agreement and the accompanying COP decisions** are focused on enhancing **efforts to mitigate and adapt to climate change beyond 2020**.
 - a. **Long Term Goal:**
 - Limiting global temperature increase well below 2 degree Celsius, while urging efforts to limit the increase to 1.5 degrees.
 - b. **Ends the Strict Differentiation between developed and developing countries:** Provides for a **framework that commits all countries** to put forward their best efforts against climate change and keep strengthening these efforts.
 - c. **Mitigation - Binding Procedural Commitments -> Preparing, communicating and maintaining NDC; Communicate new progressive NDC every five years;**
 - d. **Carbon Markets:** Though the agreement avoided any direct reference to the use of market based approaches, it recognized that the parties may use 'internationally transferred mitigation outcomes' to implement its NDCs.
 - e. **STOCKTAKE/SUCCESSIVE NDCs**
 - To ensure successive improvement in efforts, the agreement provides for **two linked processes**, each on a five-year cycle.
 - **Global Stocktake** to assess collective progress towards the agreement's goals. The first global stocktake took place in 2023.
 - **New NDCs** every five years informed by the outcomes of the global stocktake. Signatories should ensure that the new NDCs are more ambitious than the previous ones.
 - g. **Finance**
 - **Provisions for Support to poor developing countries by Developed countries.**
 - **Finance Mobilization goal**
 - The COP decided to extend the \$100 billion-a-year goal through 2025, and beyond that, by 2025 COP will set a "new collective quantified goal from a floor of "\$100 billion a year".

h. **Adaptation:**

- A major priority for many developing countries was strengthening adaptation efforts under the UNFCCC. The agreement does that by :
 - Establishing a global goal of "enhancing adaptive capacity, strengthening resilience and reducing vulnerability to climate change"
 - Committing enhanced adaptation support for developing countries
 - Including a review of adaptation progress, and of the adequacy and effectiveness of adaptation support, in the global stocktake to be undertaken every five years.

i. **Loss and Damage:**

- In a victory to small island countries and other countries highly vulnerable to climate impacts, the agreement includes a **free standing provisions** extending the Warsaw International Mechanism for Loss and Damage
 - The mechanism, established at COP-19 is charged with developing approaches to help vulnerable countries cope with unavoidable impacts, including extreme weather events such as sea-level rise.
 - Potential approaches include **early warning systems** and **Risk insurance**.
- Loss and Damage provision "did not involve or provide a basis for any liability or compensation.

- **Analysis: Positives**

- **PARIS Agreement** was a major breakthrough whose significance could be understood from the fact that with **150 Presidents and Prime Ministers**, it was the largest ever single day gathering of heads of state surpassing even the UN summits.
- **It turned the corner** after the **failure of Kyoto Protocol** and inability to reach an agreement in previous COPs. It represented a change in global attitude and recognized that climate change is a global problem which should be dealt immediately.
- The agreement is **more Comprehensive than Kyoto protocol** which was limited to assigning greenhouse gas emission targets for a group of developed countries.
- **Regular stock take** would increase the chances of world community rectifying its targets to prevent climate crisis.
- **Best Compromise possible:** Developed countries ensured climate action would be joint responsibility of every nation unlike Kyoto Protocol; Developing countries were able to take heart by the fact that all important principle of differentiation - has been retained, even though in diluted form; Island nations and least developed countries were happy to force the rest of the world to acknowledge the need to take a 1.5 degree path instead of the 2 degree it is more comfortable with.

- **Analysis: Negatives/Limitations**

- **NDCs** are not ambitious enough and the world is moving towards missing the goal of limiting the temperature rise to 1.5 degree celsius.
- **Non-Binding nature** raises a question on effectiveness of the implementation.
- **Frequent reviews** by stocktaking may be challenging to achieve as it is difficult to generate a Paris like consensus regularly.
- **Exit Clause is also problematic:** A country can exit from the agreement with one year's notice after three years have passed from the time a country ratifies the Agreement.

- Given the fractured politics in many countries withdrawal of any major emitter would lead to rapid unraveling of the agreement. (for e.g. the US had withdrawn from the agreement)
- **Financing in real terms have come down when various data shows need for increase**
- **Even after eight years of the deal, by 2023, a lot of work still needs to be done about market mechanism; increasing funding; and making NDCs more ambitious.**

4) INDIA'S UPDATED NDC UNDER PARIS AGREEMENT

- India submitted its INDC on 2nd Oct 2015.
- The NDC submitted in Aug 2022 is India's first NDC under the Paris Agreement. The Article 4, paragraph 9 of the Paris Agreement provides that each Party shall communicate a nationally determined contribution every five years in accordance with the decision of COP21.
- So, in Aug 2022, India communicated an update to its first NDC submitted earlier on Oct 2, 2015 for the period upto 2030, as under:
 - » To put forward and further propagate a healthy and sustainable way of living based on traditions and values of conservation and moderation, including through a mass movement for 'LIFE'– 'Lifestyle for Environment' as a key to combating climate change [UPDATED].
 - » To adopt a climate friendly and a cleaner path than the one followed hitherto by others at corresponding level of economic development.
 - » To reduce Emissions Intensity of its GDP by 45 percent by 2030, from 2005 level [UPDATED].
 - » To achieve about 50 percent cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030, with the help of transfer of technology and low-cost international finance including from Green Climate Fund (GCF) [UPDATED].
 - » To create an additional carbon sink of 2.5 to 3 billion tonnes of CO2 equivalent through additional forest and tree cover by 2030.
 - » To better adapt to climate change by enhancing investments in development programmes in sectors vulnerable to climate change, particularly agriculture, water resources, Himalayan region, coastal regions, health and disaster management.
 - » To mobilize domestic and new & additional funds from developed countries to implement the above mitigation and adaptation actions in view of the resource required and the resource gap.
 - » To build capacities, create domestic framework and international architecture for quick diffusion of cutting-edge climate technology in India and for joint collaborative R&D for such future technologies.
- This update to India's existing NDC is a step towards our long-term goal of reaching net-zero by 2070.

5) THE COONTINUING UNFCCC NEGOTIATION

A) COP26 (GLASGOW PACT), 2021 : KEY OUTCOMES

- **Mitigation:**
 - » It asked countries to strengthen their 2030 climate action plan or NDCs by 2022.
 - » First clear recognition of the need to move away from fossil fuels -> it called for "phase down of coal" and "phase out of inefficient fossil fuel subsidies".
- **Adaptation:**
 - » Asked developed countries to atleast double the money being provided for adaption by 2025 from the 2019 levels.

- » It created a **two year work program** to define a goal on adaptation.
- **Paris Rule Book was finalized**
 - » 'Transparency Framework' was completed - it included reporting rules and formats for emissions, progress on pledges and financial contributions.
 - » Carbon Market provisions have been finalized [a major achievement of COP26].
 - **Credit generated from earlier periods**, including through Clean Development Mechanism were transferred to the Paris Agreement but only since 2013. This will allow developing countries to meet its first NDC targets.
 - On the **issue of double counting**, it has been decided that a country that generates a credit will decide whether to authorize it for sale to other nations or to count towards their climate targets. The emission cuts will be **counted only once**.
- **Problems that remained:**
 - » Funding
 - » Loss and Damage
 - » Didn't specifically raise emission reduction targets.

B) COP27 (AT SHARM EL SHEIKH, EGYPT)

- **Quotes:**
 - » The **UN Secretary General had declared at the start of the conference**, "We are on a highway to climate hell with foot still on the accelerator".
- **Key Highlights:**
 - » Nod for establishment of Loss and Damage Fund.
 - » **Estimates of Financial Requirements -> COP27 agreement** for the first time, quantified the financial needs for climate action. It said about US\$ 4 trillion had to be invested in the renewable energy sector every year till 2030 if the 2050 target of net zero was to be achieved.

C) COP28: DUBAI, UAE

- **Practice Question:**
 - » "The COP28 Declaration has left almost all the problems where they were before" Elaborate [10 marks, 150 words]
- The meeting reviewed the Progress of commitment made by 197 countries under the Paris Agreement to mitigate the razing global warming.
- **Outcome: Dubai Consensus:**
 - » Negotiators adopt resolution **titled "Dubai Consensus"**; the text reflects a compromise between developed and developing countries on emissions.
- **Highlights of Global Stocktake (GST):**
 - » The GST text echoed the GST input findings that **1.5 degree target would require "deep, rapid and sustained" reduction in global emissions of 43% by 2030 and 60% by 2035** from the 2019 levels and **eventually reaching net zero by 2050**.
 - » **Fossil Fuel Phase-out:** COP28 agreement has called upon countries to contribute towards "transitioning away" from fossil fuels and phase down of unabated coal power so as to **achieve net zero by 2050**.
 - **Criticisms:**
 - No timelines

- Not using the phrase "fossil fuel phase-out" and instead the use of "transitioning away".
 - While calling for phase down of "unabated coal power, the door was left open for "low-carbon fuels", "low emission" technologies, "low-carbon hydrogen" - all terms with very loose definitions.
 - » **Tripling global renewable energy capacity by 2030** (from 3400 GW today to 11000 GW) and **doubling of global average rate of energy efficiency improvements by 2030**.
 - » **Accelerating and substantially reducing non-carbon-dioxide emissions globally**, including in particular methane emissions by 2030.
 - **Criticisms:** No target mentioned
 - **Note:** A group of about 100 countries at Glasgow (in 2021) had made a voluntary commitment to reduce methane emissions by 30% by 2030.
 - » **Reduction of emission from road transport** on a range of pathways, including through development of infrastructure and rapid deployment of zero-and low-emission vehicles;
 - » **Phase down of inefficient fuel subsidies** that don't address energy poverty or just transition, as soon as possible.
- **Operationalization of L&D Fund:**
 - **Background:** A decision to set up a Loss and Damage Fund had been taken last year in Sharm el-Shaikh (COP27) but it had not been created, and no money had been promised.
 - **COP28 operationalized the fund** and several countries have already made commitments worth around \$800 million by the end of the conference.
 - **This is the most significant outcome for vulnerable countries** as L&D fund is meant to provide financial help to countries trying to recover from climate-induced disasters.
 - **Santiago network** has also decided to avert, minimize, and address loss and damage to catalyze the technical assistance of relevant organizations, bodies, networks and experts for the implementation of relevant approaches associated with climate change impacts.
 - **Global Goal on Adaptation (GGA):**
 - **Background:** COP26 at Glasgow had decided to set up a two-year work program to define the contours of adaptation framework.
 - » The two-year work program resulted in **identification of some common adaptation goals** like reduction in climate-induced water scarcity, attaining climate-resilience in food and agricultural production, supplies and distribution and resilience against climate induced health impacts.
 - The COP28 retains calls for a doubling in adaptation finance and plans for assessment and monitoring of adaptation needs in the coming year.
 - » An explicit 2030 date has been integrated into the text for targets on water security, ecosystem restoration, health.
 - **Issue of Climate Finance Targets** will be reviewed in next COP:
 - Currently, the \$100 billion goal hasn't yet been met (although it appears on track this year) and is far short of what is needed.
 - **COP28 saw an agreement to draft a post 2025 finance target ahead of COP29**. This is a step forward, but details will only be hammered next year.
 - **COP28 Declaration on Climate Change and Health**

- » This is the first ever move to commit action and finance to combat the health impact of climate change.
- **Other Related Outcomes:**
 - A group of **22 countries** signed a declaration to triple nuclear energy capacity between 2020 and 2050, in order to reduce dependence on oil, gas, and coal.
 - **G7 countries** have announced to phase out coal by 2030 and have urged G20 countries to also agree on it.
 - India and Sweden co-launched Phase II of the Leadership Group for Industry Transition (LeadIT 2.0) for the period 2024-26 at COP-28. They also launched the Industry Transition Platform, which will connect the governments, industries, technology providers, researchers, and think tanks of the two countries.
 - **Green Industrialization Initiative:** African leaders came together on the third day of COP28 to launch the initiative. The GII is set to accelerate green growth of industries in Africa and attract finances and investment opportunities.
- **Limitations/Criticisms:**
 - » **Countries failed to adopt rules to set up global carbon market:** Civil society has hailed the move as parties didn't agree to adopt weak rules for carbon markets.
 - » **Climate Finance** issue is still pending and would be taken up in COP25.
 - » **No timelines for fossil fuel transitioning:** The text related to fossil fuel transitioning is weak, in-adequate and with loopholes.
 - » **NDCs** remain far away from achieving Net Zero by 2050.
 - » **Net Zero by 2050** target is expected to bring pressure on China and India whose net zero targets are for 2060 and 2070 respectively.
 - » **Major Decisions** have not been integrated with agendas like 'Common but differentiated responsibilities'.

6) LOSS AND DAMAGE FUND

- **Introduction:**
 - » COP28 of UNFCCC in Dec 2023 operationalized the loss and damage Fund. Loss and Damage Fund consist of compensation that rich and industrialized nations, with historical role in climate change, must pay to poor nations. COP28 decided that the fund will be serviced by new, dedicated and independent secretariat. It will be supervised and governed by the Board.
- **Why is it needed?**
 - » Compensate poor countries for irreversible damages due to climate change caused by rich industrialized nations.
 - » This fund can be used to slowdown loss of ecosystem services; protect food security; address climate induced socio-economic degradation; and ensure climate justice.
- **Some challenges:**
 - » Absence of consensus on activities which will be classified as loss and damage activities.
 - » Difficulty in quantifying the amount of loss and damage.
 - » Poor data availability on how loss and damage will be reported.
 - » Low pledges so far by industrialized nations

- » Poor countries may lack capacity to use the fund
- **Way forward:**
 - » Mutually agreeable definition of L&D needs to be identified on priority.
 - » Standard mechanism of quantifying damages needs to be developed.
 - » Incorporate the concept of L&D in national and international policies
 - » Steps to fight climate change - Mitigation and Adaptation - needs to be intensified.

7) INDIA'S DECISION TO ADOPT NET ZERO BY 2070

- At COP26, PM Modi has proposed a **five fold strategy** for India to play its part in helping the world get closer to 1.5 degrees Celsius. India's '**Panchamrita**' promises include:
 - India will get its **non-fossil energy capacity** to **500 GW** by 2030.
 - This is a **50 GW increase** from its existing target.
 - India will meet 50% of its energy requirements till 2030 with renewable energy.
 - India will reduce its projected carbon emission by one billion tonnes by 2030.
 - India will reduce the carbon intensity of its economy by 45 percent by 2030.
 - India will achieve net zero by 2070.
- **India's demand from developed countries:** In the spirit of climate justice, the developed countries should be providing at least \$1 trillion in climate finance to assist the developing countries and those most vulnerable.
- **Analysis:**
 - This is a **very positive move** to take Indian on firm path towards decarbonization.
 - This announcement also keeps in mind the **Common but Differentiated Responsibilities and Respective Capabilities (CBDR-RC)**.
 - India's net zero comes in 2070 and NDC is subject to funding from developed countries
 - **India is contributing more than its share:** Despite a 2070 net zero year for India, India's cumulative emissions between 1900-2100 would be lower than the US, China or EU.
 - **India continues to show international leadership**
- **Critics of shifting to a Net Zero target**
 - **Over-appropriation of global carbon budget** by a few.
 - Countries which have higher emissions presently are taking more advantages of the environment.
 - The campaign to achieve net zero by 2050 is designed to achieve Paris goals by the "lowest cost" methods, foregoing equity and climate justice.
 - **Wasn't mandated by Paris Agreement.**
 - **India is anyways a small contributor** - Our emissions are 4.37% of the world's share (with 18% population).
- **Critics of Sustainability of India's Net Zero Strategy**
 - India's plan to increase dependence on hydro projects and nuclear energy will create displacement, deforestation, hazardous radiation etc.
 - Solar and Wind Energy is also focused on Mega energy parks which may cause displacements.

- **Way Forward:**
 - **Identify short-term and medium-term targets** to achieve the long term goals.
 - **Areas of GHG Reduction:**
 - **Decarbonizing India's Energy Sector** - Replace fossils with renewables; Improve efficiency of existing fossil fuel using sectors; Remove unavoidable carbon release through Carbon Sequestration.
 - **Green Transformation of Mobility:**
 - Shift in modal mix from road to rail and fuel diversification approach to encourage sustainable fuels (biofuels, CNG, LNG) in short term.
 - Electrification in medium term.
 - Hydrogen based heavy mobility in the long term.
 - **Radical Decarbonization of Industrial sector** including steel, cement, chemicals and fertilizers.
 - Accelerate circular economy; efficiency improvement; electrification of heat; carbon capture and low carbon fuels such as biomass and hydrogen.
 - **Green Building, Infrastructure and Cities:** India's top 25 cities contribute to 15% of its estimated GHG emissions.
 - Rethink urban planning with a focus on transit oriented urban development and an emphasis on low-carbon buildings and infrastructure construction.
 - **Agriculture:** Agriculture sector is the largest contributor to nitrous oxide (N₂O) and Methane (CH₄).
 - A national campaign to empower, educate and enable more than 100 million farmers in adopting precision agriculture, sustainable animal husbandry, and green energy.
 - **Integrate emission reduction with Climate Adaptation**
 - Strengthen a suit of social protection program, especially for people facing rural distress.
 - Invest in disaster preparedness.
 - **Corporate India** has a vital role to complement government efforts. The goals of 21st century India Inc should be to foster innovative and inclusive green development.
 - **Strengthening State Capacity** can help the country move from reactive decision making to proactive planning and execution. A **Low-Carbon Development Commission** supported by the overarching framework of a climate law, could play this role.
- **Conclusion:**
 - The work towards achieving net zero by 2070 will allow India to demonstrate genuine climate leadership for the rest of the developing world, and secure a better, greener future for its citizens.

8) ISSUE OF CLIMATE FUNDING

- **Introduction**
 - » Money has been central to many a fight at the Climate Change negotiations. UNFCCC as part of its CDDR principle requires developed countries to provide financial assistance to developing nations in their fight against the climate change.
 - » **Globally**, there are two funding mechanisms - **The Green Climate Fund** and the **Global Environment Facility**.

- **Green Climate Fund (GCF)**
 - » Established at COP-16 in 2010, it is the financial mechanism for UNFCCC under article 10. It is regarded as the chief instrument for the fulfillment of developed world's annual support of \$100 billion annually till 2025.
- **Global Environment Facility (GEF)**
 - » Created at Rio Earth Summit in 1992 to help tackle planet's most important environmental problems. It is an **independent, international partnership** of 183 countries, international institutions, Civil Society Organizations and the private sector that addresses global environmental issues.
- **Current Funding Situation:**
 - » **Requirement: As per COP27** (Sharm el-Sheikh agreement), the global transition to a low-carbon economy would likely require about US\$ 4-6 trillion every year till 2050. This is 5% of the global GDP.
 - » **Availability:** The \$100 billion amount, that the developed countries have promised is the only money in play right now.
- **So key problems of current climate fundings are:**
 - » **Requisite finance** hasn't been mobilized.
 - » **Funding bias in favour of climate change mitigation activities.** This bias is there because mitigation efforts are easily visible in short run and returns from adaptation efforts will be visible after long time.
 - » **Developing world** in itself cannot fight the climate crisis as they are still struggling for finance for their development needs.
 - » **A number of countries** are unable to access global finance
 - » **Lack of transparency** is leading to problems of double counting and green washing.
- **Way Forward:**
 - » **Availability and Access** are two main dimensions to the problem of climate finance
 - » **Increasing Availability:**
 - **Developed countries** need to increase their contribution.
 - **Mobilize resources from private sector:** Businesses and Corporations need to invest money into green projects. Barely 30% of current financial flows are coming from private sources.
 - **Creation of right environment for investments in green project** -> Private sector will not invest unless they are reasonably sure of healthy returns.
 - **Carbon Tax** - Common citizens will have to contribute to the bulk of the additional financial resources.
 - » **Increasing Access:** Present rules and regulations of global financial systems, make it difficult for many countries to access international finance, particularly those with political instabilities. Therefore, there is a need to simplify lending mechanisms and overhaul credit rating systems.
 - » **Increased Transparency:**
 - Climate finance flows through a maze of channel - bilateral, regional, multilateral. It is in the form of grants, concessionary loans, debt, equity, carbon credit, and more. As a result

there are widely different opinion on the quantum of climate finance currently being mobilized. This needs to be addressed.

- **Conclusion**

- Though, more money is flowing in green economy than a few years ago, but the pace of increase is nowhere adequate. When it comes to climate change, **along with money, it is the time which is in short supply**.

9) CARBON SEQUESTRATION

- It is the process of capturing and storing atmospheric carbon dioxide. It is one of the methods of reducing the amount of carbon dioxide in the atmosphere with the aim of fighting climate change.
- **There can be two major types of carbon sequestration:**
 - **Biological:** Biological Carbon Sequestration is the storage of carbon dioxide in **vegetation** such as grasslands, or forests, as well as in **soils and oceans**.
 - **Plant rich landscapes** like forests, grasslands etc. capture 25% of the global carbon emissions.
 - **Soil** can store carbon in the form of Soil Organic Carbon. Soil can also store carbon as carbonates.
 - **Colder and nutrient rich part of ocean** can absorb more carbon dioxide than warmer parts. Therefore, polar regions generally serve as carbon sink.
 - **Geological Carbon Sequestration**
 - It is the process of storing carbon dioxide in underground geologic formations, or rocks.
 - **Naturally, Carbonates** are created over thousands of years when carbon dioxide dissolved in water and percolates in soil, combining with calcium and magnesium minerals, forming '**caliche**' in desert and arid soil.
 - **Artificially**, CO₂ captured from industrial, or any other sources may be injected into porous rocks for long-term storage.
 - **Hydrodynamic Trapping:** It refers to a time-dependent hydrogeological process where injected CO₂ is effectively trapped by the existence of very long travel times to the surface.
 - **Solubility Trap:** CO₂ dissolved in liquid like water or oil.
 - **Mineral Carbonation:** CO₂ can be made to react to naturally occurring minerals to form stable compound which can stay like that for years (e.g. Calcium carbonate)
- **Technological Carbon Sequestration**
 - These are the new ways being explored by scientists to capture and store carbon using innovative technologies and to make useful products out of it.
 - **Graphene Production**
 - **Direct Air Capture** - Capturing carbon directly from air using advanced technology plants.
 - For now the technology is highly expensive and energy intensive. But with more advancement in technologies, this may become a viable option.
 - **Engineered Molecules** - These molecules can change shape by creating new kinds of compounds capable of singling out and capturing carbon dioxide from the air.

10) GEO-ENGINEERING

- **Introduction:**
 - » **Definition:** Geo-engineering is a theoretical concept which aims to modify and cool environment to defeat the global warming. It may involve reduction of Sunlight reaching earth or absorption of CO₂ to reduce global warming (Carbon Capture Technologies).
 - » Since the global community is looking for a Net Zero target by 2050, the Geo-engineering technologies are expected to play a key role in this.
- **Reduction of sunlight reaching Earth:**
 1. **Stratospheric Aerosol Injection:** Injecting the atmosphere with Sulphur/ Hydrogen Sulphide (copies volcanic effect and scatters sunlight).
 2. **Putting Large Mirrors in Space** - reduce the amount of sunlight reaching earth.
 3. Using Wind-Powered Motors to **whiten the cloud** -> by spraying water into the sky -> reflect solar radiation
- **Carbon Capture and Storage (CCS)** (Or Carbon Capture Utilization and Storage (CCUS)) refers to technologies that can capture CO₂, at a source of emissions before it is released into atmosphere.
- **CDR** takes the form of both natural means like afforestation or reforestation, and technologies like direct air capture where machines mimic trees by absorbing CO₂ from their surrounding and storing it underground.
 - **E.g. Fake Trees** containing compounds which can react with CO₂ to absorb it and store it in solid form.
- **How significant is the role of CCS and CDR** in achieving net-zero by 2050?
 - In IPCC AR6, there is no pathway to 1.5 degrees C that doesn't use CDR.
- **Limitations/Problems with these CCS and Geoengineering method:**
 - **CCS and CDR** are still technologies under development without demonstrated feasibility at large scale despite decades of development.
 - It also suffers from other challenges like high energy requirements; high cost; challenges in the transport and long-term storage of carbon.
 - **CDR methods like afforestation, reforestation, Bioenergy with Carbon Capture and Storage (BECCS)** are constrained by their **need of land**. It may also hamper food and water security.
 - **Ocean Iron Fertilization:** The Convention of Biological Diversity has already imposed a de facto moratorium based on precautionary principle. It could result in eutrophication, which may adversely affect the ocean ecosystem.
 - **Stratospheric Aerosol Injection** is also highly controversial as this could have unintended effects on global and regional climates.
 - Further, there are concerns related to **fairness, equity, and justice** in the adoption of geo-engineering technologies as most of the R&D is dominated by North American and Western Euro.

- **Why very little progress?** - Lack of policy support and spending on R&D.
- **Geopolitics of Geoengineering:** Since, developed countries dominate the R&D and discussions on futuristic governance framework, there are concerns about the representation of positions of developing and underdeveloped countries.
- **Way forward**
 - **Improved policy support**
 - **Learn from successful implementation of technology in industrialized countries.** Increased collaboration with global industries.
 - **Strengthen international geo-engineering governance:**
 - Future use of geo-engineering should take into consideration the core principles of UNFCCC like common but differentiated responsibilities.
 - The international governance should introduce accountability, oversight and transparency into the use of geo-engineering in future. The governance framework should be inclusive in approach.

11) MISSION LIFE

- **Details about Mission LiFE**
 - It was first proposed by PM Modi at **COP 26 of UNFCCC** in Nov 2021. It is envisioned as **an India led global mass movement that will nudge individual and collective action to protect and preserve the environment**.
 - It embodies the spirit of the P3 Model: Pro Planet People.
 - It functions on the basic principles of '**Lifestyle of the planet, for the planet and by the planet**'.
 - At the launch, PM Modi also highlighted that the **concept of 'Reduce, Reuse and Recycle' and circular economy**; and mentioned that it has been part of the Indian Lifestyle for thousands of years.
 - LiFE also resonates with **climate justice** -> it highlights enhanced obligations for those in developed countries and supports climate adaptation and mitigation for those most affected and yet least responsible.
- **Significance:**
 - According to UNEP, more than 2/3rd of the GHG emissions can be attributed to household consumption and lifestyles -> therefore the urgent cuts to global emissions we need can only be achieved through widespread adoption of greener consumption habits.
 - Life recognizes that small individual actions can tip the balance in the planet's favor.
 - » Actions such as saving energy at home; cycling and using public transport instead of driving; eating more plant-based foods and wasting less; and leveraging our position as customers and employees to demand climate-based friendly choices.
 - Many of the goals of LiFE can be achieved by deploying 'nudges', gentle persuasion technique to encourage positive Behaviour.
 - » The UNEP employs proven nudging techniques:
 - Discouraging Food waste by offering smaller plates in cafeterias;

- encouraging recycling by making bin lids eye-catching;
- and encouraging cycling by creating cycle paths

12) INITIATIVES TO TAKE AHEAD THE LIFE 'LIFESTYLE FOR ENVIRONMENT'

- To take ahead LiFE movement announced by the Hon'ble PM in 2021, **MoEF&CC** has introduced **two pioneering initiatives** that indicates the country's proactive approach to climate change, sustainability, and promotion of eco-conscious practices.
- Both the programs Green Credit Program (GCP) and The Ecomark Scheme were notified in Oct 2023.

A) GREEN CREDIT PROGRAM (GCP):

- The program is a domestic voluntary market mechanism which incentivize environmental actions across diverse sectors and by different stakeholders like individuals, communities, ULBs, private sector etc.
 - » Under **environmental actions**, 8 activities have been identified. It includes Afforestation, Water Conservation, Sustainable Agriculture, Waste management, air pollution reduction, mangrove conservation and restoration, Ecomark (a government scheme to identify environment friendly products) and sustainable building and infrastructure.
- The **environmental actions will earn green credits** and these green credits will be tradable and those earning them will be able to put these credits up for sale on a proposed domestic market platform.
- **In its initial phase**, the GCP focuses on two key activities: Water Conservation and Afforestation.
- **How to obtain Green Credit:**
 - **Registration of the Activity:** To obtain Green Credits, Individuals and entities must register their activities through the central government's dedicated app/website www.moefcc-gcp.in.
 - **Verification:** The administrator will verify the activity through a designated agency, with self verification for small projects.
 - **Granting of Certificate:** After completion of the verification, the administrator will grant Green Credit Certificate which will be tradable on the green credit platform.
- **How is Green Credit Initiative different from Carbon Credit Initiative:**
 - **Carbon Credit** can be claimed by reducing carbon footprint, whereas Green Credit has a **much wider scope** and can be claimed by various types of environment friendly activities.
 - While carbon market (carbon credit) is more focused at industry and corporations, green credit program can **benefit individuals and communities**.
- **How will Green Credit work in case of afforestation:**
 - Under the initiative, registered and approved entities can pay to finance afforestation in specific tracts of degraded forest, and wasteland. The actual afforestation will be carried by the forest department. **Two years after planting** - and following an evaluation by ICFRE - each planted three could be worth one 'green credit'.
 - **How credits can be used?**
 - These credits can be bought to meet obligations under Corporate Social Responsibility (CSR) and Environmental, Social, and Governance (ESG). It can also be used to meet compensatory afforestation requirements under the Forest (Conservation) Act.

- These green credit could be used by companies which have diverted forest land and non-forest purposes and razed thousands of trees to offset some of their obligations under India's compensatory afforestation laws.
- **Weeks** after the Union Environment Ministry announced the rules for its Green Credit Program (GCP), ten states have identified parcels of degraded forests land that will be made available for individuals, groups, public and private sector units to earn - and potentially trade - green credits.

- **Concerns:**

- **Linking Green Credits to Compensatory afforestation** activities is even trickier as the program in essence facilitates the creation of land banks that could be easily diverted to commercial entities and contribute to even greater level of diversion of forest land.

B) THE ECOMARK SCHEME:

- The Scheme replaces the previous notification and provides accreditation and labelling for household and consumer products that meet specific environment criteria while maintaining quality standards as per Indian norms.
 - Products accredited under the scheme will adhere to specific environment criteria, ensuring minimal environment impact.
- It will build consumer awareness; and encourage eco-conscious choices.
- It will also motivate manufacturers to shift towards environment friendly products.
- The scheme also seeks to ensure accurate labelling and prevent misleading information about the products.
- **Administrator:** The CPCB administers the eco-mark scheme in partnership with Bureau of Indian Standards (BIS), which is the national body for standards and certification.
- **Conclusion:** Both Green Credit Program and Ecomark Scheme mark significant steps in promoting sustainable living, environmental conservation, and through individual and collective choice, embody eco-friendly practices in India. They align with global sustainability goals and reflect the government's commitment to conservation and protection of the environment.

13) INDIA AND CLIMATE CHANGE

- **Target:** NDC and Panchamrita (already discussed)
- **Steps to mitigate climate change:**
 - Updated NDC
 - Panchamrita
 - Mission Life – Green Credit Program; Ecomark Scheme
 - NAPCC
 - Strengthening of renewable energy sector (already discussed)
 - International Initiative: ISA, OSOWOG etc.
 -
- **Challenges:**
 - High dependency on coal
 - Methane emission – crucial for rice and animal husbandry
 - Increasing population and economy -> increase energy requirement
 - Various issues associated with renewable energy sector (already discussed)

- Targets not ambitious enough (as per 'Climate Action Tracker' – India's target climate and action remains "highly insufficient").

- **Way Forward:**

- Deal with constraints of renewable energy sector (already discussed with renewable energy)
- Emphasize more on Mission Life (Participation of people is crucial in fight against climate change)
- Look into indigenous knowledge for conservation and biodiversity protection
- Adapt – a situation may come where climate change is difficult to prevent -> adaptability of our food system, cities, health sector etc would be crucial.

14) INDIA'S TRADITIONAL PRACTICES IN FIGHT AGAINST CLIMATE CHANGE

- **How traditional practices in India can contribute to climate change mitigation and adaptation:**

- **Organic Farming Natural Farming:** Less use of energy and resources and less emissions
- **Sacred Groves:** Protecting forests
- **Adapting to rise in sea level:** E.g. Kuttanad Kaliyanam farming of Kerala, where cultivation happens below sea level.
- **Traditional rainwater harvesting:** E.g. Zabo in Nagaland and Jhalaras of Rajasthan. It will reduce variability of water availability.
- **Sustainable architecture:** Traditional houses made up of mud, bamboo and thatched hut need less energy for heating or cooling purposes.
- **Community based resource management** – for e.g. 'Van Panchayats' or 'Pani Panchayats' played an important role in conservation, sustainable utilization and equitable distribution of resources.

- **Way Forward:**

- Focus on identifying and documenting such traditional practices.
- **Get the above advantages scientifically verified**
- Promote these practices
- Spread awareness about the relevance of these practices.

15) GREENWASHING

- **Definition:** It is a deceptive marketing practice where companies portray a product or service as environment friendly when in reality it is not.
- **Why is greenwashing a problem?**
 - Misleads a consumer
 - Undermines genuine green products and services
 - Damage trust and discourage people from supporting genuine green products
 - Negatively impact innovation
- **Central Consumer Protection Authority (CCPA)** has released a draft guideline on Prevention and Regulation of Green Washing issued under the Consumer Protection Act 2019.
 - It defines greenwashing and prohibits it.
 - The guidelines apply to advertisers, endorsers service providers etc.

- Gives mechanism for verification of claims through certification, independent third party verification etc.
- **Other steps to prevent Greenwashing:**
 - BIS has a standard for eco-labelling of products and services
 - SEBI's Business Responsibility and Sustainability Reporting (BSRS) norms

16) CRYOSPHERE AND IMPACT OF CLIMATE CHANGE

- Cryosphere refers to earth's frozen region. They are found at very high altitude and in the polar region. It consists of snow, glaciers, ice caps etc. They play a important role in regulating climate by reflecting sunlight and also by storing carbon in the permanently frozen region. They are also important source of freshwater.
- **How is climate change impacting cryosphere?**
 - Melting of snow/ice -> rise of sea level
 - Thawing of permafrost is also leading to emission of methane.
 - Damaging biodiversity as the life in the region has evolved to survive in very cold, icy conditions
- **How can this be problematic for humans:**
 - **Water security** may be threatened.
 - **Sea level rise** will lead to submergence of coast and other issues
 - **General increase in temperature** -> climate-based devastation will intensify
 - **Ecosystem disruption** -> may lead to hampering of ecosystem services.
- **Steps being taken:**
 - Steps to fight climate change
 - Focus on Himalayan ecosystem under NAPCC
 - National Centre for Polar and Ocean Research – play a role in increasing india's contribution in international polar research program.
 - HIMANSH – a research station established in Himachal Pradesh.
 - UNESCO's World Glacier Monitoring Services

3. DESERTIFICATION AND LAND DEGRADATION

- **Introduction**
 - » Desertification is a type of land degradation in which relatively dry land region becomes increasingly arid, typically losing its bodies of water as well as vegetation and wildlife.
 - » Currently, **41% of the landmass worldwide** is prone to desertification and more than 2 billion people are affected by desertification and land degradation.
- **Scenario in India: Almost 30% (96.4 million hectare) of India's land mass is under desertification**
 - » This **desertification is further increasing** -> 26 of the 29 Indian states have reported an increase in the area undergoing desertification in the last 10 years.
 - » According the MoEF&CC, India loses, 23 hectares of dryland to drought and desertification every minute, which further leads to a loss of 20 million tonnes of potential food grain production.
- **Key causes**

- i. **Deforestation:** Decreasing forest/tree cover is the most important factor for increasing penetration of deserts in different parts of the world.
 - It also leads to **increased loss of soil cover**, mainly due to rainfall and surface runoff which is one of the biggest reasons for desertification.
 - ii. **Overgrazing and unsustainable agri practices** are other major factors leading to desertification.
 - iii. **Increasing Pollution** also negatively hampers biodiversity (including biodiversity), causes infertility of soil and promotes desertification
 - iv. **Climate Change and higher probability of droughts** have made more areas vulnerable to desertification
 - v. **Salination** caused by **overuse of water**, degrades soil and promotes desertification.
 - vi. **Unsustainable Mining practices** also degrades the geographic region and promotes desertification.
 - vii. **Invasive species** of plants such as *Prosopis Juliflora* have also resulted in the decline of natural vegetation and expansion of deserts.
 - This can be specifically seen in case of Banni Grassland, of Kutch Gujarat.
 - viii. **Forest fires** are the other major drivers of desertification.
 - ix. **Lack of Resources** to fight desertification
 - The issue was also raised recently in the 14th COP on UNCCD. Only \$6.4 billion have been spent in last 2 years to combat desertification, the real cost should be as much as \$450 billion annually.
- **Impact**
- i. **Threatens socio-economic development** by threatening food security, increasing poverty and unemployment due to land degradation.
 - ii. **Increases vulnerability of already vulnerable groups**
 - A very large section of poor people in the world live on degraded lands, desertification is especially problematic for them.
 - It also leads to reduction in food production and thus reduces availability and increases cost.
 - iii. **Promotes the vicious cycle of degradation**
 - Poverty force people to go for unsustainable agri practices, further promoting desertification.
 - iv. **Desertification adds to and worsens the impact of climate change**
 - It reduces forest cover and thus reduces the sinks for CO₂.
- **International Efforts**
- i. **United Nation Convention on Combating Desertification (UNCCD)**
 - ii. **World Day to Combat Desertification and Drought:** 17th June
 - iii. **The Bonn Challenge:** It is a global goal to bring 150 million hectares of degraded and deforested landscapes into restoration by 2020 and 350 million hectares by 2030.
 - iv. **Great Green Walls Initiative:** This initiative was **launched in 2007 by African Union** and is aimed at restoring Africa's degraded landscapes and transform millions of lives in one of the world's poorest regions, the **Sahel**. It covers areas from Senegal in the west to Djibouti in the East of Africa.

- **National Efforts**
 - » Though India **doesn't have a specific Policy or legislative framework for combating desertification**, many of our national policies have enabling provisions for addressing the problem.
 - **Various watershed development Programs**
 - **Green India Mission** is aimed at protecting restoring and enhancing India's diminishing forest cover and responding to climate change.
 - **Green Highway** initiative of MoRT&H is also aimed at planting trees, besides the national highway throughout the country.
 - » **Desertification and Land Degradation Atlas of India**
 - » **Pilot Project to Restore Degraded Forest Landscapes**
- **Way Forward**
 - » **Fast track all the efforts regarding afforestation, sustainable agriculture, organic farming, agro-forestry etc.**
 - » **Poverty eradication programs** should be integrated with **strategies to fight desertification.**
 - » The **soil and land preservation** efforts should be prioritized and mainstreamed.
 - » **International Cooperation through UNCCD** has to increase to ensure more funds, more focus on vulnerable countries.
 - Funds for climate change mitigation and adaptation could be linked to the anti-desertification program.
 - » **Secure land rights** for indigenous communities who have lived sustainably with nature for centuries.
 - » **Promote community led initiatives** like making degraded lands out of bounds for grazing, growing fast-growing plants and tall trees that serve as barrier against winds and sandstorms.
- **Conclusion**
 - » Desertification is as big a challenge as climate change and has the potential to impact 2 billion of the world's population by 2040. It's high time that all the stakeholders give fighting desertification the required focus.

4. OCEAN ACIDIFICATION

- Ocean acidification is the ongoing **decrease in the pH of the Earth's oceans**, caused by the absorption of carbon dioxide (CO₂) from the atmosphere. When CO₂ is absorbed by seawater, it reacts with water molecules to form carbonic acid, which increases the acidity of the water.
- **What is the PH of Ocean?**
 - The Current pH of the ocean is around 8.1, which is slightly alkaline (basic) on the pH scale. However, this pH value is decreasing over time due to ocean acidification.
- **Key Factors behind Ocean Acidification:**
 - **Increasing CO₂** in the atmosphere -> increased absorption of CO₂ in water
 - **Eutrophication:** When dead algae is decomposed -> CO₂ is produced.
 - **Land Use Changes:** Deforestation and other land use changes can also contribute to ocean acidification by reducing the ability of plants to absorb CO₂ from the atmosphere
 - **Agriculture:** Fertilizers and other chemicals can also contribute to ocean acidification by increasing the amount of CO₂ in the atmosphere.

- **Acid Rain:** It has impact on surface ocean chemistry and has major impact locally and regionally but very small impact globally.
- **Effects of Ocean Acidification:**
 - **Impaired Shell formation:** Many marine organisms, such as corals, mollusks, and some planktonic species, rely on calcium carbonate to build their shells and skeletons. However, increased acidity can make it more difficult for these organisms to form and maintain their shells, which can have cascading effects throughout the food web.
 - **Alteration in Biodiversity:** Some species may be able to adapt to the changing conditions, while others may be more vulnerable to the effects of increased acidity
 - **Disrupted Food Web:** For example, if the growth and reproduction of certain species of plankton are impaired, it can reduce the amount of food available for larger organisms that feed on them
 - **Decreased Resilience:** Some studies have suggested that increased acidity can make marine organisms more susceptible to other stressors, such as pollution or disease, which can further impact their survival and growth
- **Steps to fight ocean acidification:**
 - Reducing CO₂ emissions, nutrient pollution, and air pollution

5. OZONE LAYER

- **Questions**
 - » Discuss the key international initiatives which helped humankind win the fight over Ozone layer depletion in last 3 decades. [10 marks, 150 words]
 - » Success of the Montreal Protocol on Ozone Depleting Substances carries a number of important lessons for fight against climate change. Elaborate [15 marks, 250 words]
- **Ozone Layer** The ozone layer or ozone shield is a region of Earth's stratosphere that absorbs most of the sun's ultraviolet (UV) radiation. It contains high concentration of Ozone (O₃) in relation to other parts of the atmosphere, although still small in relation to other gases in the stratosphere.
 - » Ozone layer is mainly found in lower stratosphere (approx. 20-30 km above earth)
- **Usefulness of ozone layer:** Prevents damaging Ultraviolet from reaching earth thus benefitting both plants and animals; Protects oxygen of lower layer which would break up by the action of ultraviolet rays otherwise.
- **Ozone Layer Depletion:** In 1970s scientists discovered that Chlorofluorocarbons (CFCs), broke apart in the atmosphere and released **chlorine atoms**. This caused the ozone depletion. The same effect resulted when bromine atoms were released by halons. Thus, **CFCs and halons** are examples of Ozone depleting substances.
- **What are the uses of Ozone depleting substances/ when and why they are produced?**
 - Chlorofluorocarbons (CFCs); Bromine containing compounds; Carbon Tetrachloride; Methyl Chloroform; Trichloroethane; Hydrochlorofluorocarbons (HCFCs); N₂O etc.
- **Extent of maximum damage:** In 2000, the areas of Antarctic Ozone hole reached a record of 29 million sq km.
- **Why is the Ozone Depletion predominant at the Antarctic (not over Arctic and other areas which produce ozone depleting substances)**

1. **Antarctic is more cold than arctic** : The Antarctic stratosphere is **much colder**. The low temperature enable the formation of PSCs, below 20 km.
2. **Stability of Vortex is longer here**

- **Environmental Effect of Ozone Depletion: Impact of UV-B radiation on living and non-living things on earth**

- » Decrease in the quantity of total-column ozone tend to cause increased penetration of solar UV-B radiation (290-315 nm) to the earth's surface. It has profound effect on human health, animal plants, microorganisms, material and air quality.
- » **Effect on Human and Animal Health**
 - Eye disease, skin cancer and infectious morbidity
 - In susceptible (light skinned colored) population UV-B radiations is the key risk factor for development of non-melanoma skin cancer (NMSC).
- » **Effects on terrestrial plants and Aquatic Ecosystem:** Physiological and developmental process are affected
- » **Effects on biogeochemical cycles:** Alternates both source and sinks of greenhouse and chemically important trace gases
- » **Effects on air quality**
 - **Higher photo dissociation rates of key trace gases** that controls the chemical reactivity of the troposphere.
 - Increase both production and destruction of ozone (O₃) and related oxidants such as hydrogen peroxide (H₂O₂), which are known to have adverse effect on human health, terrestrial plants, and outdoor materials.
 - Can lead to increased production of particulates such as cloud condensation nuclei.
- » **Effects on Materials**
 - Synthetically occurring polymers and naturally occurring bio-polymers as well as other materials are adversely affected by solar UV radiation.
 - It increases photodegradation of these materials, limiting their life outdoors.

- **International Initiatives to contain the depletion of the Ozone layer**

1. **Vienna Convention:** Signed in 1985 and came into force in 1988. The objective of the convention were for countries to promote cooperation by means of systematic observations, research and information exchange.
2. **Montreal Protocol** : Signed in Sep 1987. The Montreal Protocol is an international treaty designed to protect the ozone layer. It came into force in 1989.
 - It aims at reducing the production and consumption of Ozone Depleting Substances (ODS) in order to protect earth's fragile ozone layer.
 - **Impact of Montreal Protocol:** It has also been a highly successful international arrangement, as it has phased-out more than 98% of the ODS which was part of its main mandate by 2021. The remaining ODS are HCFCs which are in the process of being phased out.
 - **What has India done under the Montreal Protocol**
 - India has already phased out CFCs, and CTC.

- **Currently** India is engaged in the phase-out of production and consumption of other Hydrochlorofluorocarbons (HCFCs) with an accelerated phase out schedule as per the Montreal Protocol.

- **Lessons from success of Montreal Protocol**

- Shows that nations of the world can successfully work together to face a common danger
- Adherence to the 'Precautionary Principle' was important because waiting for complete scientific proof could delay action to the point where the damage will be irreversible.
- Principle of CBDR (common but differentiated responsibility) can be successful.
- Consistent and credible signals to industry is important. It gives them an incentive to develop new and cost-effective alternative technology.

3. Kigali Agreement - Amendment to Montreal Protocol

- **About Kigali Agreement to Montreal Protocol.**

- Kigali agreement refers to an amendment to the 1989 Montreal Protocol to eliminate planet-warming HFC gases.
 - It calls for phasing-out of HFCs, a set of 19 gases in Hydrofluorocarbon family that are used extensively in air-conditioning and refrigerant industry.
 - These gases are not ozone depleting but are thousands of times more dangerous than carbon dioxide in causing global warming.
 - **Why put the target in Montreal Protocol and not UNFCCC**
 - Montreal Protocol is much more successful than the UNFCCC and have fairly good track record in controlling various kinds of emissions.

- **Present Situation: Scientific Assessment of Ozone Depletion**

- Action taken under the MP have led to **long term decrease in the atmospheric abundance of controlled ODS** and the **ongoing recovery of stratospheric ozone**.
- Ozone layer** in parts of stratosphere has recovered at a rate of 1-3% per decade since 2000. At this rate, the northern hemisphere and mid latitude ozone is schedule to **completely heal** by the 2030s followed by southern hemisphere in the 2050s and the polar region by 2060.
- Full compliance of Kigali agreement** can reduce future global warming due to HFCs by about 50% between now and 2050 compared to a scenario without any HFC controls.
 - **World can avoid upto 0.4 degree of global warming** this century through **implementation of the Kigali Amendment**
- Important steps that can be taken to fasten ozone recovery:**
 - **Complete elimination** of controlled and uncontrolled emissions of substances such as Carbon tetrachloride and dichloromethane.
 - **Bank recapture and destruction of CFCs** halons, and other HCFCs.
 - **Elimination of HCFC and methyl bromide production.**
 - **Mitigation of Nitrous oxide emission.**

- **Conclusion**

- The **depletion of ozone layer is consistently been contained**, thanks to global efforts to ban the use of harmful chemicals that destroy ozone.
- In Sep 2019, the UNEP has said that the ozone layer over some areas in northern hemisphere will be completely restored in their pre-1980 levels by as early as the 2030s. The Antarctic ozone hole could be completely healed by the 2060s.
- The Montreal Protocol now **should also act as a model of cooperation among global community to fight Climate Change.**

6. SOME NATIONAL EFFORTS IN BIODIVERSITY CONSERVATION

1) THE BIOLOGICAL DIVERSITY (AMENDMENT) ACT, 2023

- It was introduced in LS by MoEF&CC in Dec 2021 and was finally passed in the house in July 2023 and Rajya Sabha in Aug 2023.
- The amendment intends to encourage the Indian system of medicine; facilitate fast-tracking of research, patent application process; attract more foreign investment in the preservation and commercial utilization of India's biological resources; and decriminalizes all the offences under the act.
- **Key Highlights:**
 - i. **Simplify compliance requirements for domestic companies**
 - ii. **Exempts AYUSH practitioners, local people, and communities of the area**, including growers and cultivators of biodiversity, from giving prior intimation to State Biodiversity Boards for accessing biological resources for commercial utilization.
 - iii. **Users of codified traditional knowledge and AYUSH practitioners will be exempted from sharing benefits with local communities**
 - **Issue/Criticism:** The term Codified Traditional Knowledge hasn't been defined by the bill or by CBD. A broad interpretation might exempt all local traditional knowledge from benefit sharing requirements.
 - Legal experts also feel that exemption to Ayush practitioners would be detrimental to ecology and go against the principle of sharing commercial benefits with indigenous communities.
 - iv. Removes research and bio-survey from the purview of benefit sharing.
 - v. Benefit sharing will be based on terms agreed between the user and the local management committee represented by the National Authority
 - **Issue/Criticism:** The bill removes direct role of local communities in determining the benefit sharing provisions.
 - vi. Companies registered in India and controlled by Indians are now treated as Indian companies, even if they have foreign equity or partnership, thereby reducing the restrictions on them.
 - vii. **Simplifying the IPR Process:**
 - The act specifies that approval of NBA is required before applying for IPR involving biological resources obtained from India, or (ii) sealing of patent.
 - The bill says that approval would be required before the approval of IPR instead of before the application itself.
 - It further differentiates between foreign and domestic entities.

- Foreign entities will require approval from NBA whereas domestic entities will be required to register with NBA. However, at the time of commercialization of IPR, domestic entities will need approval from NBA.
 - viii. The bill **decriminalizes all offences** under the act and provides for wide range on penalties. Further, it authorizes government officials to hold enquiries and determine penalties.
 - **Issue/Criticism:** Such discretion with government official may promote corruption/rent seeking.
 - ix. The bill allows for foreign investment in research into biodiversity. However, this investment will necessarily have to be made through Indian companies involved in biodiversity research.
 - x. **A new section - 36(A)** has been added emphasizing on the monitoring of the Biological Resources obtained from foreign countries for use in India as per the provisions of the Nagoya Protocol on access to benefit sharing.
 - xi. **Section 36(B)** enables state government to develop strategies and plans for conservation and sustainable use of biological diversity.
- **Other criticisms:**
- **Lack of Focus on Biodiversity Conservation:** the bill prioritize intellectual property and commercial trade at the expense of the key aim of conserving biological diversity.
- **Positives:**
- The bill encourages a conducive environment for investments - by promoting Ease of Doing Business
 - Simplification of patent process would encourage innovation in India and ensure more IPR for companies in India.
- **The report by the joint parliamentary committee** examining Biological Diversity Amendment Bill 2021, was finally submitted in Parliament in Aug 2022. **Key Recommendations:**
 - **Promoting Indigenous Medicines:**
 - Reducing the pressure on wild medicinal plants by encouraging cultivation of medicinal plants.
 - Indian system of medicine should be encouraged by clearly defining codified traditional knowledge.
 - Promote indigenous research
 - **Exemption to Ayush Practitioners:**
 - The committee clarified that Ayush practitioners who have been practicing indigenous medicine, including Indian system of medicine, as a profession, for sustenance and livelihood are exempt from prior intimation to state biodiversity boards to access biological resources.
 - **FDI Inflows:**
 - There is a need to attract more foreign investments in the chain of biological resources, including research, patent, and commercial utilization, without compromising national interest by defining foreign companies as per the companies act and by defining a protocol for utilization of biological resources from India.
 - The committee recommended that **biodiversity management committee** under the proposed law and indigenous communities should be **empowered by clearly defining benefit claimers** to be conservers of biological resources.

- Offences should attract civil penalties with proportionate fine structure so that the violators don't escape. It shouldn't be too meagre.

2) FOREST CONSERVATION ACT, 1980 AND 2023 AMENDMENT

- **Why in news?**
 - The Forest (Conservation) Amendment Act 2023, came into force from 1st Dec 2023.
- **Introduction**
 - It was enacted for the conservation of forests by restricting de-reservation of forest and forest land for non-forest purposes.
- **Supreme Court Verdict and Expansion of the Conservation Function:**
 - In 1996, the **Supreme Court** in a verdict in **TN Godavarman Thirumulpad vs Union of India** had expanded the definition and scope of forest land to include all areas recorded as forest in government record, irrespective of ownership, recognition and classification.
- **Forest Conservation (Amendment) Act, 2023**
 - Changing the name to **Van (Sanrakshan Evam Samvardhan) Adhiniyam**, i.e. **Forest (Conservation and Augmentation) Act**.
 - The amendment provides that **two types of land** will be under the purview of the act:
 - i. Land declared/notified as a forest under Indian Forest Act, 1927 or under any other law, or
 - ii. Land notified as a forest on or after Oct 25, 1980 in a government record.
 - **Exempted Category of Land:**
 - **Under the 1927 Act**, decisions regarding diversion of forest land for non-forest purposes are taken by the state government. **The 1980 act** requires additional prior approval from central government.
 - **The amendment** provides that such approval will not be required when forest land is diverted for constructing
 - i. **Strategic Linear Projects** (roads, railways) of national importance and concerning national security within 100 km of India's border.
 - ii. **Security related infrastructure** upto 10 hectares,
 - iii. **Defence related projects**, a camp for paramilitary forces, or public utility projects as specified by the central government, not exceeding five hectares in a left-wing extremism affected area.
 - The bill also exempts **certain type of land** from the provisions of the act, such as forest land along a rail line or a public road maintenance by the government providing access to a habitation, or to a rail, and roadside amenities upto a maximum size of 0.10 hectare.
 - **Assignment/Leasing of forest land:**
 - The Original act restricts the de-reservation of forests for use of forest and **non-forest purposes**. Such restrictions can be lifted only with the prior approval of central government.

- The act specifies certain activities that will be excluded from non-forest purposes, meaning that restrictions on the use of forest land for non-forest purposes will not apply.
- The **amendment adds** more activities to the list such as (i) zoos and safaris under WPA, 1972 owned by the government or any authority, in forest areas other than protected areas, (ii) ecotourism facilities, (iii) silviculture operations (enhancing forest growth), and (iv) any other purpose specified by central government.
- **Significance:**
 - **Increase Plantation in non-forest areas:** The environment ministry says that the application of the act on land covered under dictionary meaning of forests (or deemed forests) has resulted in a "declining tendency in plantations in non-forest lands owing to the apprehension among individuals, organizations, and authorities regarding such plantations being considered forests".
 - **Removal of mandatory central government approval** for diversion of forests in certain cases is expected to reduce delays in the implementation of strategically important projects.
- **Criticism:**
 - The amendment by providing blanket exemptions from the act for several types of lands and several types of projects is going **contrary to the intent of the 1980 act** which was enacted for prevention of de-reservation of forest lands and large scale deforestation.
 - Such forest clearance activities will also violate rights of forest dwellers which were secured under the 2006 Act.
 - The **Apex Court in *T.N. Godavarman Thirumulkpad v. Union of India (UOI) and Ors* (AIR 1997 SC 1228)**, has given a wider meaning to forest land to broaden the scope of their protection. However, the **2023 bill provides a much narrower interpretation** to the term forest land by inserting section 1A to encompass only two types of land under its ambit.
 - **Exemption near border area** would cover large parts of the north-eastern region, Uttarakhand and Himachal Pradesh and could lead to loss of biodiversity in biodiversity hotspots.
 - **The purpose of allowing a zoo** inside a forest is not clear.
 - There is also a **lack of clarity on compensatory afforestation** for this diverted land.
- **Conclusion:** The overall tenor of the proposal is to make it a little easier to use forest land for non-forestry purpose. The idea is to balance the development and security needs with the forest conservation needs of India. But while doing so, it should be ensured that long term forest conservation and rights of tribals and forest dwellers are not affected in long run.

3) FOREST COVER AND FOREST CONSERVATION

- **Current Situation of forests in India: As per Indian State of Forest Report 2021**, India's total forest cover is 7.13 lakh sq km (21.71% of India's total area). When compared to 2011, there has been an increase of 3.14% in the total area under forest.
- **Concerns:**

- **Decline in Natural Forests:** What is worrying is that 1,582 sq km decline has taken place in moderately dense forests, or "natural forests".
- **Decline in North-East India:** Five states in Northeast - Arunachal Pradesh, Manipur, Meghalaya, Mizoram, and Nagaland have all shown loss in forest cover. Its important for protecting the forest cover of NE as it is natural forest and with only 7.98% of the geographical area, it contributes to 23.75% of India's total forest cover.
 - **Reasons:**
 - A spate of natural calamities; particularly landslides and heavy rains
 - Anthropogenic factors: Shifting agriculture, pressure of development activities and felling of trees.
- **Vulnerability to Forest Fire:** Around 35.46% of the forest cover in India is prone to forest fires.
 - Between Nov 2020 to June 2021, 3.4 lakh forest fire hotspots were recorded, which is the highest recorded forest fire cases ever.
- **Why forests are crucial? Why is it important to protect forests:**
 - **Controlling Global Warming:** Forest restoration will play a huge role in achieving Net Zero climate target. A study in 2017 showed that land-based sinks (natural climate solutions which also includes forests) can provide upto 37% of emission reductions.
 - **Protecting Biodiversity**
 - **Preventing Desertification**
 - **Ensuring Soil Health** (by preventing soil erosion, increasing organic content of soil etc.)
 - **Protecting micro-climate** of an area
 - **Livelihood** for crores of tribals and forest dwellers
 - **Water Security:** Forest restoration is must for water security.
 - **Medicinal plants** from forests can solve a number of health issues
- **Key Initiatives in India to Conserve Forests:**
 - Forest Conservation Act, 1980
 - Compensatory Afforestation Regime
 - Various Protected Areas - NP, WLS, Tiger Reserves, Biosphere reserves etc.
 - **Green India Mission** launched in 2010 with three objectives:
 - Double the area to be taken up for afforestation/ eco-restoration in India in the next 10 years.
 - Increase the green house removal by India's forests
 - Enhance the resilience of forests/ecosystem.
 - **Nagar Van Scheme**
 - **Forest Fire Prevention and Management Scheme**
 - **India's International commitments:**
 - India has committed to restore 5 million hectares of degraded and deforested land between 2021 and 2030.
 - India also targets creation of 2.5 to 3 billion tonnes of carbon sink by 2030 as part of its INDC. This would require India to increase its tree cover by 12% over the next 10 years.
- **Key International Initiatives:**
 - **REDD+** under UNFCCC framework:

- **New York Declaration on Forests (NYDF)** in 2014 by United Nations Climate Summit: Target of restoring 350 million hectares of forests by 2030.
- **The span 2021-2030** is the UN Decade on Ecosystem restoration. It emphasizes on efforts to restore degraded terrestrial ecosystems including forests.
- **Bon Challenge** launched in 2011 focuses on global goal of restoring 150 million hectares of degraded and deforested landscapes by 2020 and 350 million hectares by 2030.
- **UN** has proclaimed 21st March as the International Day of Forests in 2012 to celebrate and raise awareness of the importance of forests.
 - The year 2022 marks a decade of IDF.
- **Key Problems of Forest Restoration in India**
 - Definition of forest - Canopy cover shouldn't define forest. Forest is something bigger.
 - Poor identification of area for restoration
 - Lack of focus on R&D strategies in tree planting
 - Stakeholder's conflicts of interest, and financing.
- **Way Forward:**
 - **Redefine Forests:** A more thoughtful definition will help India design and plan effective interventions to protect, restore and conserve a forest.
 - **Discourage diversion of forest for non-forest activities:**
 - Forests are complex ecosystem that are built over the years due to interplay of birds, mammals, reptiles, insects, amphibians, fungi, micro-organisms, water, soil, environmental conditions etc. They can't be replaced by just tree cover.
 - Compensatory afforestation relies on notions that forests are replaceable and it has led to diversion of several hundred hectares of forest cover only in 2020.
 - **Effective interventions for forest conservation:**
 - **Afforestation programs** should move away from monoculture and non indigenous species. Monoculture plantations lacks the biodiversity or ecological value, diversity of quality of forests.
 - **Scientific evidence based methodology** should be used to determine right type of tree based interventions most suitable for certain land-use.
 - Policy framework should be focused on curtailing deforestation, while improving the ecology and biodiversity of a landscape.
 - **Focus on forest landscape restoration:** It is the process of regaining ecological functionality and improving human welfare across deforested or degraded landscape. A crucial aspect of this process is to ensure the diversity of species while planting trees.
 - **Ensure flow of funds,** duly providing for the adequate participation of local people in planning and implementation through local institutions.
 - **Empower forest dwellers and tribal communities:** Enabling communities to design forest restoration by extending rights to manage forest areas promote more inclusive environmental governance.
 - Farmer managed natural generation (FMNR) systems where local communities protect and manage the growth of trees that regenerate naturally have proven to deliver several economic and ecosystem benefits.